GreenMind AI: A Multi-Metric Forecasting and Decision-Support Platform for Cloud GreenOps

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***Abstract*—Modern cloud operations require optimizing resource utilization, cost, and environmental footprint simultaneously. GreenMind AI is an open-source decision-support prototype integrating five one-hour-ahead gradient-boosted forecasters, a five-domain multi-agent analysis engine (Cost, Performance, Sustainability, Security, Reliability), deterministic conflict arbitration, a non-destructive Digital Twin simulator, and an eight-tool explainable AI copilot. Evaluation on a deterministic 60-day synthetic dataset (5,756 records; 80/20 chronological split: N_train=4,604, N_test=1,152) shows MAE improvements of 25.4%-45.4% over a one-hour persistence baseline. Our audit reveals the stored rolling baseline reflects a 15-min prediction rather than the 1-hour target; the true four-step rolling baseline yields higher error than persistence (CPU: 7.263% vs. 6.500%) due to diurnal phase shift. All evaluations derive from synthetic generator equations; live remediation is guarded by HTTP 501. Synthetic accuracy must not be conflated with verified production savings.**

***Index Terms*—cloud computing; workload forecasting; gradient boosting; carbon-aware computing; multi-agent systems; digital twin; GreenOps.**

I. INTRODUCTION

Cloud data centers consume a substantial and growing share of global electricity, driving GreenOps and FinOps disciplines [4, 6]. Proactive workload forecasting supports right-sizing, autoscaling, and carbon-aware scheduling, but high historical accuracy does not guarantee realized savings in production systems [3, 8]. Existing platforms treat cost, performance, sustainability, and reliability as siloed concerns. GreenMind AI addresses this by unifying: (i) multi-metric gradient boosted forecasting, (ii) five-domain multi-agent orchestration via a state graph, (iii) deterministic conflict resolution with explicit audit logs, (iv) non-destructive Digital Twin simulation, and (v) an eight-tool AI copilot grounded in live application state.

Contributions:

1. End-to-end architecture linking telemetry ingestion, multi-metric forecasting, multi-agent arbitration, Digital Twin simulation, and decision-support interfaces.
2. Verified forecasting performance on five targets with a baseline alignment audit identifying a 15-min vs. 1-hour horizon discrepancy in the stored rolling baseline.
3. Explicit empirical scope boundaries: synthetic data provenance, unverified live connectors, and the HTTP 501 live-remediation safety guard.

II. RELATED WORK

Calheiros et al. [7] showed ARIMA-based proactive scaling reduces QoS violations. Friedman's Gradient Boosting Machine [1] constructs additive tree models suited to structured lag features. Rossi et al. [3] evaluated probabilistic LSTMs on real cluster traces, showing point forecasts miss operational uncertainty and accuracy degrades under cross-provider transfer. Bergmeir et al. [2] established chronological holdout as standard practice. Henderson et al. [4] and Lannelongue et al. [5] formalized ML energy and carbon reporting standards. Radovanovic et al. [6] deployed carbon-aware scheduling at Google datacenters using day-ahead grid predictions. Dodge et al. [8] measured operational carbon intensity across cloud instances. GreenMind AI extends these with a unified prototype evaluated on synthetic telemetry.

III. PLATFORM ARCHITECTURE AND IMPLEMENTATION

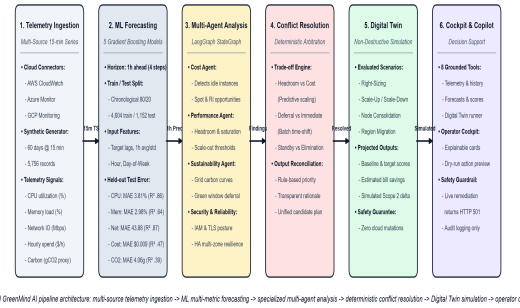


Fig. 1. End-to-end GreenMind AI pipeline architecture: multi-source telemetry ingestion -> ML multi-metric forecasting -> specialized multi-agent analysis -> deterministic conflict resolution -> Digital Twin simulation -> operator decision support.

A. Telemetry and Service Layer

FastAPI (Python 3.12; SQLite/PostgreSQL) ingests five 15-min signals: CPU (%), memory (%), network (Mbps), cost (USD/h), carbon proxy (gCO2/15-min). AWS CloudWatch (boto3), Azure Monitor REST, GCP Cloud Monitoring v3 connectors fall back to synthetic generator when credentials absent.

B. Multi-Metric ML Predictor

Five GradientBoostingRegressor models: 150 estimators, depth 3, lr=0.08, seed 42. Backward-looking features: current value y_t, hour, day, rolling mean mu_{t,4}, rolling sigma_{t,4}. Cost: CPU, memory, cost, hour, day. Carbon: grid intensity (gCO2/kWh), hour, day, CPU. Anomaly flag: z-score > 2.5.

C. Multi-Agent Analysis Engine

Five agents in a LangGraph StateGraph: CostAgent (right-sizing, Spot/RI), PerformanceAgent (saturation, autoscaling), SustainabilityAgent (green windows 01:00-06:00 UTC), SecurityAgent (IAM, TLS, ports), ReliabilityAgent (multi-AZ, backup RPO). Graceful sequential fallback with ID deduplication when LangGraph unavailable.

D. Deterministic Conflict Resolution

Three arbitration rules: (i) Performance vs. Cost (35% <= CPU < 75%): right-size + predictive autoscaling, 60-s buffer, confidence 0.88; (ii) Reliability vs. Cost (CPU < 35%): downsize + automated failover, confidence 0.92; (iii) Carbon deferral vs. Execution: shift non-critical batches to green windows, confidence 0.87. Each emits a full audit record.

E. Digital Twin Simulator

Five non-destructive scenarios: Right-Size, Scale-Up/Down, Consolidate, Region Migration (e.g., in-north ~650 gCO2/kWh to ca-central ~90 gCO2/kWh). Recomputes Optimization Score (0-100) before/after. All outputs: is_simulated=True, label='SIMULATED / ESTIMATED'.

F. AI Copilot and Remediation Boundary

Eight grounded tools: get_cloud_metrics, get_history, get_predictions, get_cost_analysis, get_carbon_analysis, get_recommendations, get_cloud_health, run_digital_twin_simulation. Non-dry-run and rollback endpoints return HTTP 501 Not Implemented.

IV. EXPERIMENTAL METHODOLOGY

A. Synthetic Dataset

generate_dataset.py: 5,760 observations at 15-min intervals from 2026-06-01, seed 42. CPU = clip(35 + 30*sin^2((h-7)/24*2*pi)*W_t + tau_t + epsilon_t, 2, 98), W_t=0.7 weekends, tau_t in [0,5]%, epsilon_t~N(0,16). Memory: [0,8]% drift. Network: 300+400*sin^2(.) Mbps. Cost: $0.192*(1+cpu/100*0.5) + N(0,0.0001) [m5.xlarge baseline].

B. Carbon Target

us-east grid intensity I(h_t) in 320-470 gCO2/kWh. Server power P=0.35 kW, step dt=0.25 h. C_t = clip(I(h_t)*P*dt + N(0,25), 0, 500) [gCO2/15-min]. Excludes dynamic PUE, server power states, Scope 3 emissions.

C. Chronological Split and Baselines

Targets shifted delta=4 steps; 4 trailing rows dropped -> 5,756 aligned records. 80/20 chronological split: N_train=4,604 (48d), N_test=1,152 (12d). No shuffling. Persistence: y-hat_{t+4}=y_t. Stored Roll-4: evaluates y-hat_{t+1}=mu_{t,4} on unshifted test (N=1,148) - a 15-min prediction. True Roll-4 from origin: yields higher error than persistence (CPU: 7.263% vs. 6.500%) due to diurnal phase lag. Metrics: MAE, RMSE, R2, Gain(%)=100*(1-MAE_model/MAE_persist).

V. EXPERIMENTAL RESULTS

Table I reports verified held-out metrics from re-running train.py (seed 42, 6.4 s). Gradient boosting outperforms persistence on all five targets: MAE gains 25.4% (memory) to 45.4% (network). CPU/network achieve R2=0.855/0.871, capturing sinusoidal generator patterns. Cost (R2=0.474) and carbon (R2=0.391) are noise-limited. Feature importances: CPU lag 69.4%, hour 22.3% for CPU; grid intensity 71.1%, hour 20.0% for carbon.

Table I. *Verified Test-Set Metrics (N=1,152). True Roll-4 yields higher error than persistence due to diurnal phase lag.*

Target	MAE	RMSE	R2	Persist.	Roll-4 (Stored)	Roll-4 (True)	Gain%
Util. (%)	3.809	4.811	0.855	6.500	4.622	7.263	41.4
ory Util. (%)	2.979	3.704	0.636	3.996	3.025	4.153	25.4
raffic (Mbps)	43.876	55.328	0.871	80.346	56.727	90.816	45.4
y Cost ($/h)	0.009	0.011	0.474	0.013	0.010	N/A	29.2
on (gCO2/15m)	4.061	5.087	0.391	5.846	4.571	N/A	30.5

VI. DISCUSSION AND LIMITATIONS

A. Synthetic vs. Real Generalization. High R2 reflects recovery of smooth sinusoidal generator formulas. Real cluster traces (Google Borg, Alibaba, Azure Functions) exhibit heavy-tailed distributions, preemptions, and non-stationary shifts absent from the synthetic series [3]. Reported metrics establish synthetic benchmark performance only.

B. Documented Limitations.

1. Data provenance: single generator configuration (seed 42, 60-day, us-east).
2. Recursive multi-step inconsistency: forecast_multi_step applies 5-min steps on 1-h models; feeds carbon output (~30-40 gCO2/15-min) into carbon_intensity trained on ~350-450 gCO2/kWh (~10x mismatch).
3. Live connectors: fall back to synthetic data; end-to-end live polling not validated.
4. HTTP 501 live remediation: all cloud mutations blocked; Digital Twin savings are projections only.
5. Carbon proxy: fixed 350 W, excludes dynamic power states, PUE, Scope 3 [5].

VII. CONCLUSION

GreenMind AI couples five gradient boosted forecasters with five-domain multi-agent orchestration, deterministic conflict arbitration, Digital Twin simulation, and an eight-tool AI copilot. On a 60-day synthetic benchmark, MAE improves 25.4%-45.4% over persistence. These results reflect synthetic-generator learning. Production validation requires open cluster traces (Google Borg, Alibaba, Azure Functions), real-time grid feeds (Electricity Maps, WattTime), recursive horizon refactoring, and end-to-end canary cloud testing.

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